Unemployment

Mbak 9515: Economics

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Readings

- Chapter 10. Unemployment
  - Measuring the unemployment
  - The three types of unemployment
  - Shortcomings of unemployment
  - Characteristics of labor market
  - Application of supply and demand

Measuring Unemployment

- Goal 3: Low unemployment
  - jobs are secure and relatively easy to find a job
  - **Bureau of Labor Statistics (BLS)** randomly calls 60,000 households
  - Each person in those households who is $\underbrace{\textit{16 years or older}}_{=\textit{working-age}}$ is placed in one of three categories:
  - **Employed**: if someone worked *full-time or part-time* during the past week
  - **Unemployed**: if someone *did not work* during the preceding week but *made some effort to find work* in the past four weeks
  - **Out of the labor force**: if someone *did not work* in the past week and *did not look for work* in the past four weeks

Measuring Unemployment

- Unemployment
  - Labor force: the total number of employed and unemployed people in the economy

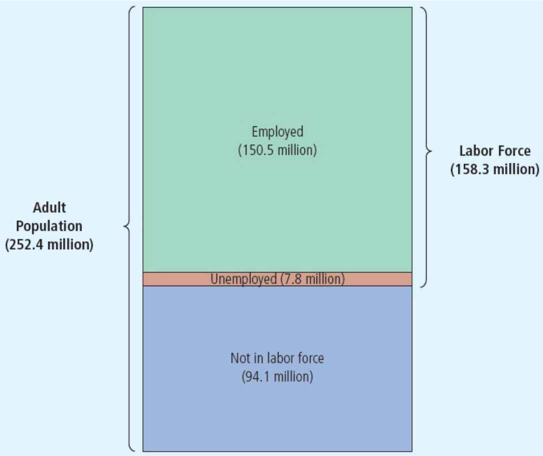
$$\text{UE Rate} = \frac{\#\text{ of Unemployed}}{\#\text{ of Employed} + \#\text{ of Unemployed}} \times 100$$

$$= \frac{\#\text{ of Unemployed}}{\text{Labor Force}} \times 100$$

- Participation rate
  - the percentage of the working-age (=adult) population in the labor force

$$\text{Participation Rate} = \frac{\text{Labor Force}}{\text{Adult population}} \times 100$$

Measuring Unemployment



Measuring Unemployment

Example

U.S. Employment Data (in millions)	
Employed	153.51
Unemployed	6.98
Not in labor force	94.66

- Labor force: $153.51 + 6.98 = 160.49$
- Working-age (adult) population: $160.49 + 94.66 = 255.15$
- Unemployment rate: $\frac{6.98}{160.49} \times 100 = 4.3\%$
- Participation rate: $\frac{160.49}{255.15} \times 100 = 62.9\%$

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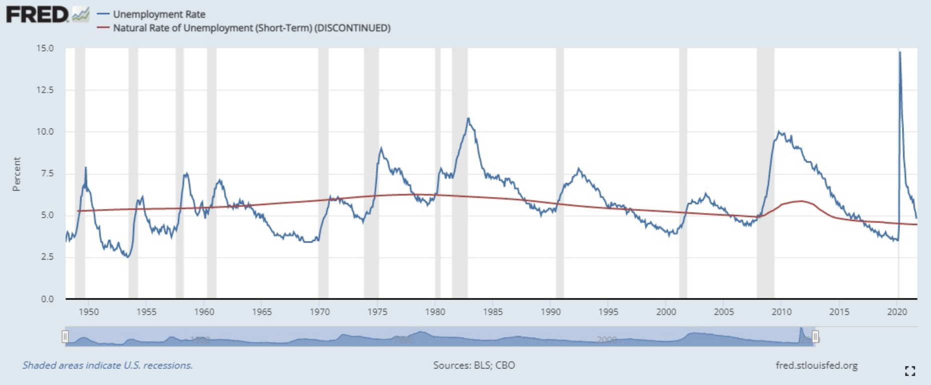
The Types of Unemployment

- Three types of unemployment: frictional, structural, and cyclical unemployment
  - **Frictional Unemployment**: the *short-term* unemployment associated with the process of matching workers with jobs
    - ★ e.g.) McDonald's introduces the "Quadstack"
    - ★ Burger King sales are hurt and BK workers are laid off
    - ★ McD's is hiring! But there *is a delay* in getting BK workers hired on at McD's
    - ★ The costs of frictional unemployment are *low* and may even be negative
  - **Structural Unemployment**: the *long-term* and chronic unemployment that exists even when the economy is producing at a normal rate
    - ★ e.g.) In 1900s, many jobs at the U.S. steel industry ⇒ fewer jobs
    - ★ New jobs at? computer industry; Apple and/or Google
    - ★ Long-term problem for some people, so the costs are *much higher* than those of frictional unemployment
    - ★ But *not* indicative of a poor economy

The Types of Unemployment

- **Cyclical Unemployment**: the extra unemployment that occurs during periods of recession
  - e.g.) US Car Industry in 2009
    - ★ High gas prices+gas-guzzling cars+housing market crash=permanent job loss
    - ★ Cyclical unemployment is representative of problem in the entire economy
    - ★ The costs of cyclical unemployment are *quite high* as it reduces real GDP significantly

Measuring Unemployment



Shortcomings of the Unemployment Rate

- Two big problems
  - Involuntary part-time workers
  - Discouraged workers
- Can we account for these shortcomings?

Shortcomings of the Unemployment Rate

Involuntary part-time workers

- Involuntary part-time workers
  - Many workers want a full -time job, but only have a part-time job
  - All involuntary part-time workers would all like to work *twice as much* and are considered
  - "half unemployed"
  - Unemployment rate with IPT Workers

$$\frac{\#\text{ of Unemployed} + \frac{1}{2}\,(\#\text{ of IPT Workers})}{\text{Labor Force}} \times 100$$

- e.g.) You learn that of the 153.51 million employed workers, 12 million are involuntary part-time workers
- Unemployment rate: $\frac{6.98}{160.49} \times 100 = 4.3\%$
- Unemployment rate with IPT Workers: $\frac{6.98+6}{160.49} \times 100 = 8.1\%$

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Shortcomings of the Unemployment Rate

Discouraged workers

- A discouraged worker is a person of legal employment age who is *not actively seeking* employment or who does not find employment after long-term unemployment
  - Discouraged Unemployment Rate

$$\frac{\#\text{ of Unemployed} + \#\text{ of Discouraged}}{\#\text{ of Employed} + \#\text{ of Unemployed} + \#\text{ of Discouraged}} \times 100$$

- e.g.) You learn that there are 5 million individuals available for work, but not currently working
- Unemployment rate: $\frac{6.98}{160.49} \times 100 = 4.3\%$
- Discouraged Unemployment Rate: $\frac{6.98+5}{160.49+5} \times 100 = 7.2\%$

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Labor Market

Demand of Labor

- Who wants to hire?
  - Producers! (firms)
- e.g.) Production and marginal product for Banana Computers Co. (Computer price=$3,000)

# of workers	Computers	MP	VMP
0			
1			
2			
3			
4			
5			
6			
7			

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5	105	17	$51,000
6	120	15	$45,000
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- How many workers should BCC hire if the going wage is \$60,000?
  - Hiring strategy: VMP$\geq$wage then hire
  - If VMP<wage $\Longrightarrow$ Cost is greater than revenue
  - So, BCC will hire 3 workers
  - If the going wage is \$50,000? $\Longrightarrow$ five workers

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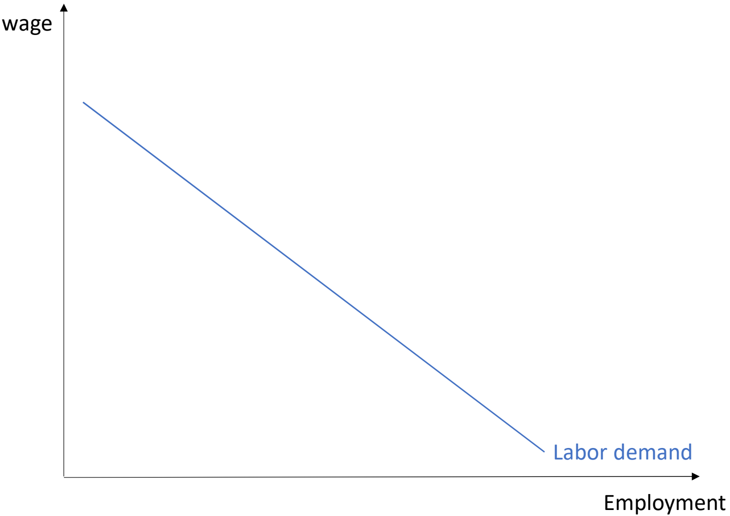
Demand of Labor



Labor Market

Shifts in the Demand for Labor

- Computer price is increased to $5,000 each

# of workers	Computers	MP	VMP
0	0		
1	25	25	$125,000
2	48	23	$115,000
3	69	21	$105,000
4	88	19	$95,000
5	105	17	$85,000
6	120	15	$75,000
7	133	13	$65,000
8	144	11	$55,000

- How many workers should BCC hire if the going wage is $60,000?
  - 7 workers!

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- How many workers should BCC hire if the going wage is $60,000?
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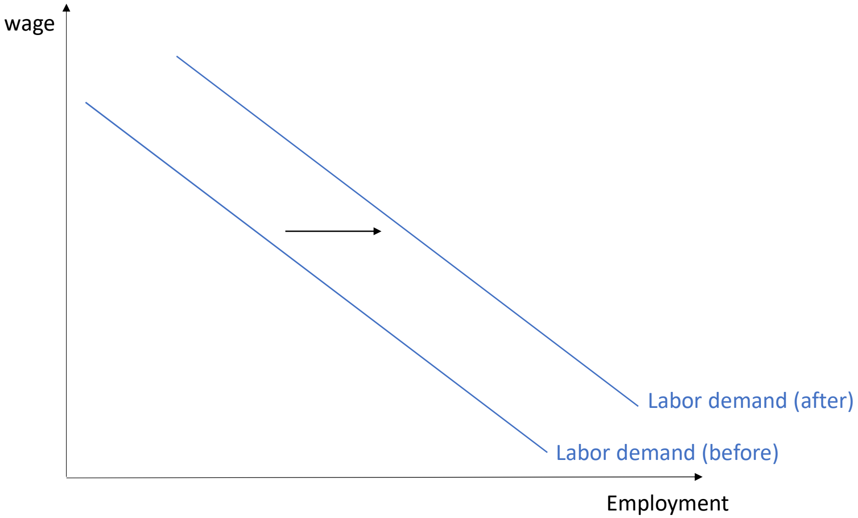
Labor Market

Shifts in the Demand for Labor



Labor Market

Shifts in the Demand for Labor

- Productivity improvements

# of workers	Computers	MP	VMP
0	0		
1	37.5	37.5	$112,500
2	72	34.5	$103,500
3	103.5	31.5	$94,500
4	132	28.5	$85,500
5	157.5	25.5	$76,500
6	180	22.5	$67,500
7	199.5	19.5	$58,500
8	216	16.5	$49,500

- How many workers should BCC hire if the going wage is $60,000?
  - 6 workers!

Labor Market

Shifts in the Demand for Labor

- Productivity improvements

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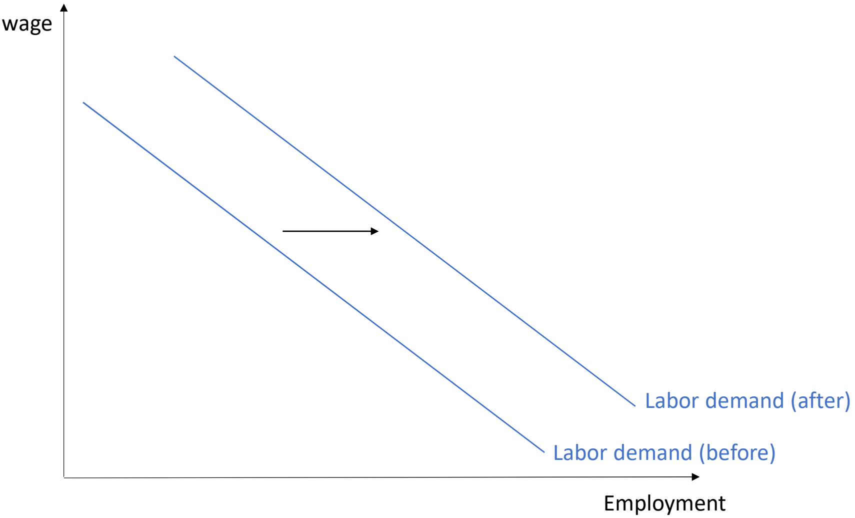


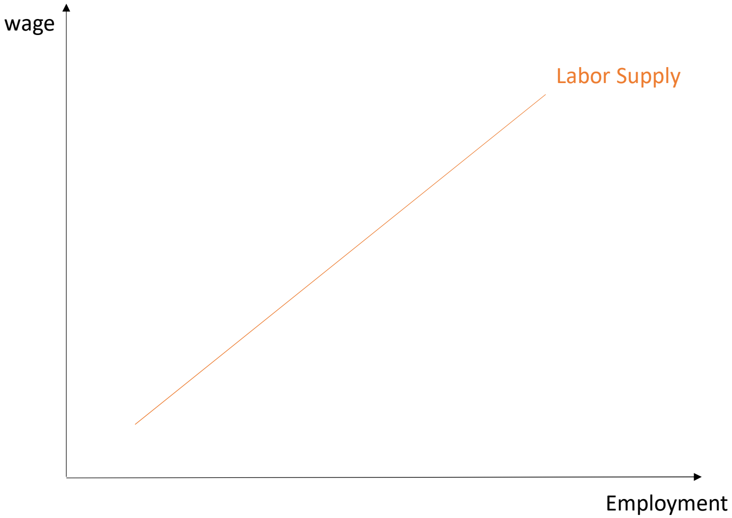
Labor Market

Supply of Labor

- Who supplies labor?
  - workers and potential workers
- Will you clean your neighbor's basement?
  - It depends on how much your neighbor will pay
  - If it is \$10?
  - If it is \$500?
  - **Reservation wage**: the compensation level that leaves you just *indifferent* between working and not working
- Your willingness to supply labor *is greater* the higher the wage

Labor Market

Supply of Labor



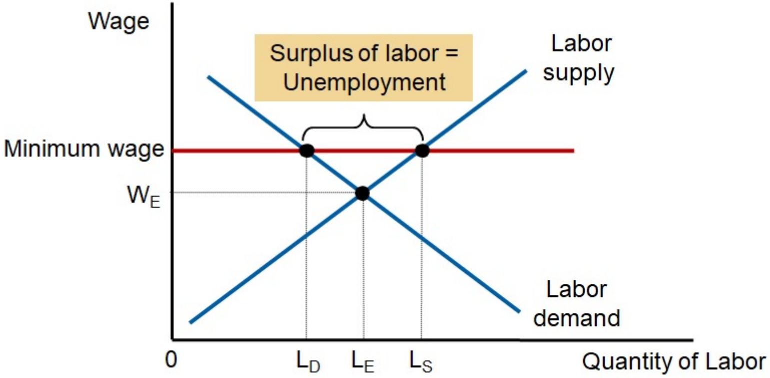
Labor Market

Shifts in the Supply for Labor

- The size of the working-age population
  - the domestic birthrate
  - immigration rate
  - the ages at which people normally first enter and retire

Minimum-Wage Laws

- Minimum-wage laws
  - Can cause unemployment
  - Forces the wage to remain above the equilibrium level



Unions & Collective Bargaining

- Union
  - Worker association
  - Bargains with employers over
    - Wages, benefits, and working conditions
  - Only 11% of U.S. workers today
    - About 33% in the 1940s and 1950s
  - Type of cartel

Unions & Collective Bargaining

- Collective bargaining
  - Process by which unions and firms agree on the terms of employment
- Strike
  - Organized withdrawal of labor from a firm by a union
  - Reduces production, sales, and profit
- Union workers
  - Earn 10-20% more than similar workers who do not belong to unions

Unions & Collective Bargaining

- Union – raises the wage above the equilibrium level
  - Higher quantity of labor supplied
  - Smaller quantity of labor demanded
  - Unemployment
  - Better off: employed workers (insiders)
  - Worse off: unemployed (outsiders)
    - May stay unemployed
    - Take jobs in firms that are not unionized

Unions & Collective Bargaining

- Are unions good or bad for the economy?
- Critics
  - Unions – a type of cartel
  - Inefficient – high union wages reduce employment in unionized firms below the efficient level
  - Inequitable – some workers benefit at the expense of other workers
- Advocates
  - Unions – necessary antidote to the market power of the firms that hire workers
  - In the absence of a union, firms pay lower wages and offer worse working conditions
  - Unions – help firms respond efficiently to workers' concerns

Theory of Efficiency Wages

- Efficiency wages
  - Above-equilibrium wages paid by firms to increase worker productivity
    - ★ Worker health
    - ★ Worker turnover
    - ★ Worker quality
    - ★ Worker effort

In Sum,

- Wages may be kept above equilibrium level
  - Minimum-wage laws
  - Unions
  - Efficiency wages
- If the wage is kept above the equilibrium level
  - Result: unemployment